Effects Of The 2022 Ukraine War On Electricity Flow Networks In Central And Eastern Europe

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Abstract

The 2022 Russian invasion of Ukraine caused major disruptions in European energy systems and affected cross-border electricity exchange patterns. This study analyzes how the war influenced electricity flow networks in Central, Eastern, and Southern Europe using network science methods. Electricity exchange data from the ENTSO-E Transparency Platform covering 2019–2024 was used to construct directed weighted networks representing cross-border electricity flows. Comparisons between pre-war (2019–2021) and post-war (2022–2024) periods were conducted using centrality measures, dependency analysis, and community detection techniques. The results show substantial structural changes after 2022, particularly in Eastern Europe. Countries such as Romania, Moldova, and Slovakia increased in strategic importance as transit and export hubs, while electricity corridors shifted toward routes connected to Ukraine. At the same time, Ukraine became more integrated into the European electricity network and increasingly dependent on electricity imports. Overall, the findings demonstrate how geopolitical disruptions can reshape interconnected infrastructure systems and highlight the value of network analysis for studying energy resilience and structural adaptation.

Introduction

Problem Definition, Motivation, Background and Hypotheses According to the European Commission (1), European electricity systems form a highly interconnected infrastructure network in which countries continuously exchange electricity across national borders. These cross-border flows are essential to balance supply and demand, stabilize power grids, reduce costs, and improve energy security. Due to this strong interdependence, disruptions in one part of the system can propagate through the broader network and influence regional electricity dynamics.

The 2022 Russian invasion of Ukraine triggered major political and economic changes in Europe, particularly in the energy sector. Reductions in Russian energy exports, changes in energy policies, market uncertainty, and increasing concerns about energy security forced European countries to adapt their electricity production and exchange strategies. As a result, electricity trade relationships and regional dependencies within the European electricity system may have changed significantly after the start of the war (2).

From a network science perspective, electricity exchange systems can be modeled as directed weighted graphs, where nodes represent ENTSO-E (3) bidding zones rather than countries, and edges represent electricity flows between them. Bidding zones are used instead of national borders because they reflect the operational structure of the European electricity market, which is shaped by transmission constraints and market design rather than political boundaries. This provides a more accurate representation of actual electricity trading and flow patterns. This representation enables the analysis of structural properties such as centrality, dependency, connectivity, and clustering. Changes in these properties can reveal how geopolitical events influence the organization and resilience of critical infrastructure networks (4, 5).

In this project, we analyze a selected subset of ENTSO-E bidding zones (3) in Central, Eastern, and Southern Europe. The selection is motivated by their structural position within the European electricity network, which makes them particularly relevant for capturing potential spillover effects of the 2022 energy shock on regional electricity exchange patterns. Using cross-border electricity flow data from these bidding zones, we construct and compare network representations for periods before and after the 2022 invasion of Ukraine. The analysis focuses on changes in node centrality, import and export dependency, shortest-path characteristics, and regional clustering patterns.

Based on the expected impact of geopolitical instability and energy market restructuring, we formulate the following hypotheses:

- **H1:** Countries geographically closer to Ukraine increased in network centrality after 2022, reflecting a shift in key transit roles within the electricity flow network.

- **H2:** After 2022 the electricity flow corridors in Central and Eastern Europe were restructured, with a shift from traditional Central European exchange routes toward corridors directly connected to Ukraine and its neighboring states.

- **H3:** Regional community structures within the network changed after 2022.

The main motivation of this work is to better understand how large-scale geopolitical disruptions affect interconnected infrastructure systems. Identifying changes in network structure and dependency relationships can provide insight into regional energy resilience, the strategic importance of specific countries within the network, and the evolving organization of European electricity exchange systems. In addition, this project demonstrates how network analysis methods can be applied to real-world infrastructure and energy-related problems.

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Related Work

Research on European electricity systems provides both useful context and methods for studying how major events, such as the 2022 Ukraine war, affect network structures. Existing work can be grouped into three main areas: regional electricity markets, network-based analyzes, and studies on dependency and resilience.

A. Regional Electricity Markets. Several papers focus on the structure of regional electricity sectors. For example, a study on South East Europe shows that while countries are strongly interconnected and depend on each other for security of supply, their markets are still fragmented due to differences in regulation and market design (6). More broadly, it has been shown that many European countries, especially smaller ones, rely heavily on electricity imports, which increases their dependence on neighboring systems (7). This highlights how important cross-border trade already was before recent geopolitical changes.

B. Network-Based Analysis. Another important line of work uses network analysis to study electricity trade. These approaches model countries as nodes and electricity flows as links, making it possible to identify key players in the system. For instance, some countries act as important "bridges" in the network, meaning they are crucial for maintaining connectivity even if their total trade volume is not the highest (8). Other studies show that electricity trade networks have become more interconnected over time, with Europe standing out as a particularly dense and well-developed region (9). A more general perspective is provided by the "network of networks" approach, which looks at how national systems interact as part of a larger global structure (10).

C. Studies on Dependency and Resilience. Finally, several studies look at dependency and resilience. Recent work highlights that the 2022 Ukraine war significantly changed EU energy trade patterns, forcing cou ntries to reduce their reliance on Russia and find alternative partners (11). To better understand vulnerabilities, Dey et al. (12) suggest going beyond global metrics and instead analyzing smaller structural patterns in the network, known as motifs, which can reveal hidden weaknesses. In addition, there is evidence that increasing domestic renewable energy production can help reduce dependence on imports (13). Similar ideas have also been applied in other energy markets, such as oil trade, to study how dependencies change after major shocks (14).
Overall, these studies show that European electricity networks are highly interconnected but also sensitive to disruptions. They also demonstrate that network-based methods are well suited to analyzing changes in structure, dependency, and resilience. Building on this, the present work examines how the 2022 Ukraine war has influenced the European electricity trade network.

Methodology

D. Data. This study uses electricity cross-border exchange data from the ENTSO-E Transparency Platform (3), which provides high-resolution information on physical electricity flows between European transmission system operators. The data set includes hourly cross-border electricity exchanges between bidding zones in Central, Eastern, and Southern Europe. A complete list of the selected bidding zones can be found in Table 1. In addition to the selected bidding zones the data set includes additional connected regions such as IT-North, which appear due to their role as import or transit partners in cross-border electricity exchanges. Ukraine is represented in the ENTSO-E dataset through multiple bidding zones reflecting its internal grid structure. In this study, we focus exclusively on Integrated Power System of Ukraine (UA-IPS), as it constitutes the main synchronized operational area and the dominant component of the reporting of the cross-border exchange of Ukraine. Excluding the alternative UA representation avoids duplication and ensures consistency in the network representation.

Region	Bidding Zones
Central and Eastern Europe	AL, AM, AZ, BA, BG, CZ, GE, HR, HU, MD, ME, MK, PL, RO, RS, RU_KGD, SK, SI, UA_IPS, XK
Southern Europe	GR

Table 1. Bidding zones included in the study and their regional grouping according to EU vocabulary (15).

From a temporal perspective, the data set covers the period from 2019 to 2024, allowing a balanced comparison of electricity exchange patterns before and after the 2022 invasion of Ukraine. This time window is chosen to capture both pre-existing structural conditions in the European electricity system and potential medium-term adjustments following the geopolitical shock. To ensure comparability, the data is divided into two periods: a pre-war baseline (2019–2021) and a post-war adaptation phase (2022–2024). This aggregation reduces the influence of short-term volatility in electricity markets and allows the analysis to focus on persistent structural changes in network topology and flow dependencies.
The ENTSO-E cross-border exchange data was filtered to include only the selected bidding zones and aggregated into annual electricity flows (terawatt-hours (TWh)) between each pair of zones. These aggregated flows were used to construct a weighted network with directed edges representing electricity transfers. The data set was then split into pre- and post-2022 periods to enable comparative network analysis.

E. Metrics. To analyze structural changes in cross-border electricity exchange, the electricity system is modeled as a directed weighted graph $G = (V, E)$, where nodes represent countries or bidding zones and directed edges represent electricity flows between them. Edge weights correspond to the total annual electricity exchange volume measured in TWh. For each year between 2019 and 2024, a separate yearly network snapshot was constructed from aggregated ENTSO-E cross-border exchange data. To analyze shortest paths and corridor importance, edge distances were defined as the inverse of electricity flow volume:

$$d_{ij} = \frac{1}{w_{ij}}$$

where w_{ij} denotes the electricity flow from country i to country j. This formulation ensures that stronger electricity exchanges correspond to shorter effective network distances and therefore represent preferred transmission corridors in shortest-path calculations. All network measures, including centrality and community detection, follow standard graph-theoretic definitions in network science (4, 5).

E.1. Centrality Analysis. To evaluate changes in the strategic importance of countries within the electricity network (**H1**), multiple node centrality measures were computed. Weighted in-degree and weighted out-degree were used to quantify total electricity imports and exports for each country. Betweenness centrality was calculated to identify countries acting as important transit hubs within the electricity flow network:

$$C_B(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}}$$

where σ_{st} denotes the number of shortest paths between nodes s and t, and $\sigma_{st}(v)$ the number of those paths passing through node v. In addition, closeness centrality was used to measure how efficiently countries are connected to the rest of the network.

E.2. Dependency and Corridor Analysis. To evaluate changes in electricity dependencies and cross-border exchange routes (**H2**), weighted import and export flows were compared across the pre-war and post-war periods. Particular attention was given to Ukraine and neighboring Eastern European countries. Additionally, edge betweenness analysis was used to identify strategically important electricity corridors and to examine how dominant exchange routes shifted after 2022.

E.3. Clustering and Community Structure. To investigate regional clustering effects within the electricity network (**H3**), community detection and clustering analysis were performed. The average clustering coefficient was used to measure the tendency of neighboring countries to form tightly connected regional exchange structures. Furthermore, Louvain community detection was applied to identify regional groupings of countries within the electricity network and to analyze how these communities changed after 2022. Connectivity properties such as shortest-path distances and the largest connected component were additionally examined to characterize overall network integration and structural cohesion.

Results

The analysis of cross-border electricity flow networks between 2019 and 2024 reveals substantial structural changes following 2022, particularly in Eastern and Southeastern Europe. Figure 1 compares the aggregated network structure for the prewar period (2019–2021) and the post-war period (2022–2024). Nodes represent ENTSO-E bidding zones, while directed edges represent annual electricity transfers (TWh), with color intensity indicating flow magnitude and edge width proportional to total exchanged volume.

A major change can be observed in Ukraine's electricity dependencies. Before 2022, Russia and Slovakia were among the largest electricity suppliers to Ukraine. Between 2022 and 2024, Moldova became the dominant electricity import partner followed by Slovakia and Hungary. At the same time, Ukraine's total electricity imports increased significantly from 4.81 TWh in 2019 to 18.64 TWh in 2024, whereas exports slightly decreased, indicating a growing dependency on external electricity supply.

Several Eastern European countries like Bulgaria, Moldova and Romania showed strong increases in betweenness centrality. In addition, weighted out-degree values, meaning the overall export of electricity, increased substantially for Moldova, Poland, and Slovakia, suggesting an expanding role as electricity exporters and transit countries.

The edge-betweenness analysis also revealed a shift in strategically important electricity corridors. Between 2019 and 2021, the most important corridors were primarily located in Central Europe, including SK→HU (Slovakia-Hungary) and PL→SK (Poland-Slovakia). After 2022, corridors directly connected to Ukraine became increasingly important, particularly RO→MD (Romania-Moldova), MD→UA-IPS (Moldova-Ukraine), and UA-IPS→PL (Ukraine-Poland).

Table 2 shows that Closeness centrality increased considerably for several countries in Eastern Europe. The strongest increases were observed for Romania, Moldova, Serbia, Bulgaria and Ukraine, indicating shorter effective network distances and stronger integration into the overall electricity network.

Table 2. Changes in closeness (C) and betweenness (B) centrality between 2019 and 2024 for selected Eastern European countries.

Country	C 2019	C 2024	B 2019	B 2024
Bulgaria (BG)	0.26	0.72	0.10	0.27
Moldova (MD)	0.17	0.61	0.00	0.19
Romania (RO)	0.28	0.80	0.10	0.33
Ukraine (UA-IPS)	0.05	0.54	0.006	0.22
Serbia (RS)	0.35	0.76	0.14	0.27

Community detection further showed changes in regional clustering structures. Before 2022, Ukraine was mainly grouped together with post-Soviet countries such as Belarus, Russia, and Moldova. After 2022, Ukraine became increasingly clustered with Central and Eastern European countries, including Poland, Slovakia, Romania, and Hungary, reflecting a structural shift toward the European electricity network.

Discussion

The structural changes observed in the network are clearly reflected in Figure 1, which shows a marked reconfiguration of cross-border electricity flows after 2022. In particular, the visual shift of dominant corridors toward Eastern and Southeastern Europe supports **H1** in showing that geographically closer countries to Ukraine increased in network importance after 2022. Multiple centrality measures confirm this pattern and indicate a growing strategic role of Eastern European countries within the electricity flow network. The substantial increase in betweenness centrality (Table 2) for Romania, Moldova, and Bulgaria suggests that these countries became increasingly important transit hubs connecting Ukraine with the broader European electricity system. Similarly, the increase in weighted out-degree for Poland, Slovakia, Moldova and new edges from Slovakia and Hungary to Ukraine indicates expanding export and redistribution functions in the post-2022 network.

The changes visible in Figure 1 and the observed shift in edge betweenness are also consistent with a reorganization of key electricity corridors after the invasion, supporting **H2**. Before 2022, the most important corridors were concentrated in tradi-

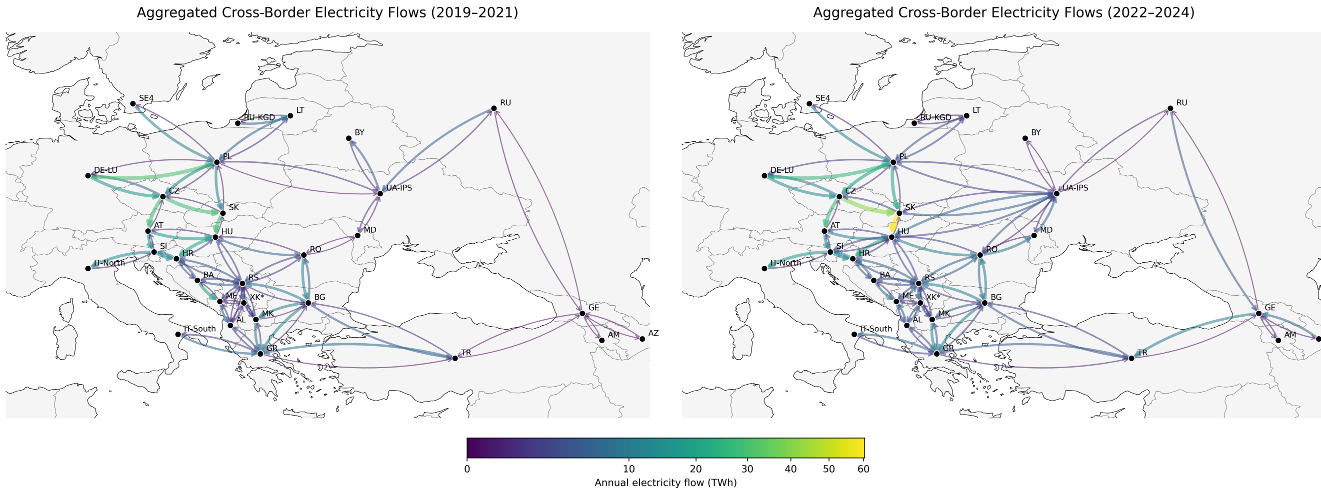

Fig. 1. Comparison of dominant cross-border electricity corridors before and after 2022.

tional Central European exchange routes. After 2022, strategically relevant corridors increasingly shifted toward routes directly connected to Ukraine, particularly through Moldova and Romania. This pattern is consistent with a structural reorientation of electricity flows and the growing integration of Ukraine into the European electricity market following the synchronization with ENTSO-E (16).

The increase in closeness centrality across Eastern and Southeastern Europe further suggests that the network became more interconnected after 2022. Ukraine's substantial increase in closeness centrality indicates that it became less peripheral within the network and more efficiently connected to other European countries. Similar developments in Romania, Moldova, Serbia, and Bulgaria point toward a broader regional restructuring of electricity connectivity.

The community detection analysis partially supports **H3**. While the network did not necessarily become more fragmented as seen in Figure 1, the composition of regional communities changed significantly after 2022. Ukraine shifted from clusters dominated by post-Soviet countries toward communities centered around Eastern and Central European states. This suggests a geopolitical and infrastructural reorientation of the Ukrainian electricity system away from Russia and toward the continental European grid.

Additionally, the results indicate that Ukraine became increasingly dependent on electricity imports after 2022. While Ukraine had previously maintained a more balanced position between electricity imports and exports, the post-2022 network shows a substantial increase in imports combined with slightly decreasing exports. This development is consistent with the severe damage to the Ukrainian energy infrastructure during the war. Russian missile and drone attacks systematically targeted electricity infrastructure, destroying or occupying large parts of the country's generating capacity. As a result, Ukraine increasingly relied on electricity imports from neighboring European countries to stabilize domestic supply (17).

Overall, the findings indicate that the Russia–Ukraine war coincided with a substantial structural reorganization of the European electricity flow network. The results consistently point toward stronger integration of Ukraine into the European electricity system and an increasing strategic importance of Eastern European transit countries.

Conclusion

This study examined how the 2022 Ukraine war affected cross-border electricity flow networks in Central and Eastern Europe using network analysis methods. The results show that the war coincided with significant structural changes in the European electricity system, particularly through the growing importance of Eastern European transit countries and the increasing integration of Ukraine into the European electricity network.

The analysis demonstrated that network-based methods are useful for identifying changes in connectivity, dependency, and regional organization within critical infrastructure systems. In particular, the observed shifts in electricity corridors and community structures highlight how geopolitical events can rapidly influence the structure of interconnected energy networks.

Future work could extend this analysis by including additional European regions, renewable energy production data, or more detailed temporal modeling to better capture short-term market dynamics and long-term structural developments.

Source Code

The source code to reproduce results presented in this paper is available at https://github.com/CorneliusNI/ul-fri-network-analysis-course-project-2025-2026.

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